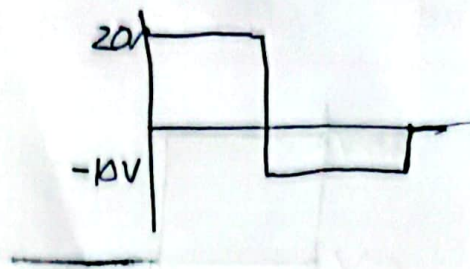
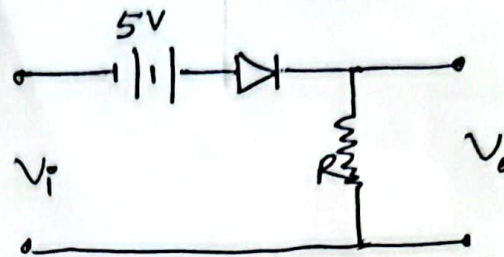


Input Waveform:



Clipper Circuit:



Working Principle:

Positive Half Cycle: The diode is in forward bias,
 $\therefore$ short circuit,

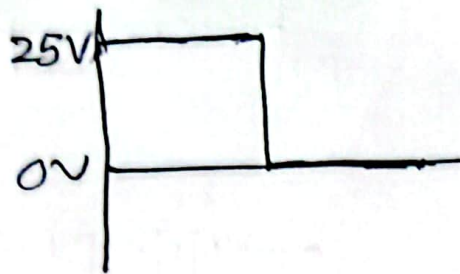
$$\therefore -V_i - 5V + V_o = 0$$

$$\therefore V_o = 20 + 5 \\ = 25V$$

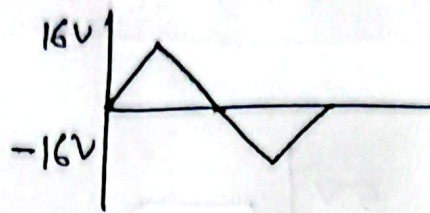
Negative Half Cycle: The diode is in Reverse bias, Means open circuit,
 No current or voltage flow,

$$\therefore V_o = 0V$$

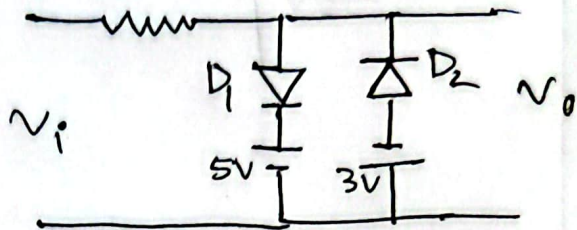
Output waveform:



Input Waveform:



Clipper Circuit:



Working Principle:

Positive Half-Cycle: D_1 forward, D_2 Reverse

$$\therefore V_o - 5V = 0$$

$$\therefore V_o = 5V$$

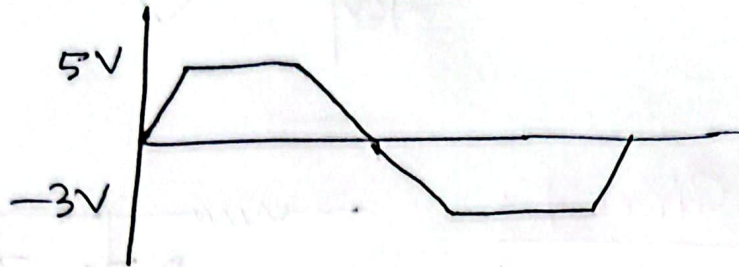
Negative Half-Cycle: D_1 Reverse and D_2 is in Forward bias,

$$\therefore V_o + 3V = 0$$

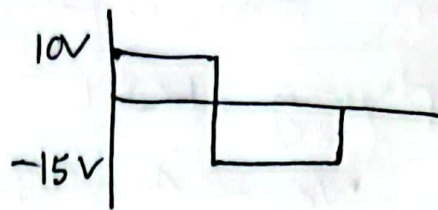
$$\therefore V_o = -3V$$

Now,

Output Waveform:

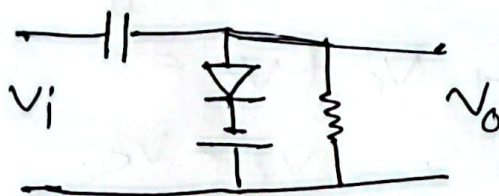


Input Waveform:



$$f = 1200 \text{ Hz}, \quad R = 75 \text{ k}\Omega, \quad C = 10 \text{ }\mu\text{F}$$

Clamper Circuit:



Working Principle:

we know,

$$\tau = RC = 75 \times 10^3 \times 10 \times 10^{-6} = 0.75 \text{ s}$$

$$T = \frac{1}{f} = \frac{1}{1200} \Rightarrow \frac{T}{2} = \frac{1}{2 \times 1200} = 0.0004 \text{ s}$$

$$\therefore \tau = RC \gg \frac{T}{2}$$

$\therefore$ this is a very good clamper circuit

When Charging: the circuit is in forward bias in positive half cycle.

$$V_o + 4 = 0$$

$$\therefore V_o = -4 \text{ V}$$

for capacitor Voltage:

Applying KVL,

$$V_c - 4 - V_i = 0$$

$$\therefore V_c = 4 + 10 = 14V$$

When Discharging: The circuit is in Reverse bias in Negative Half-cycle.

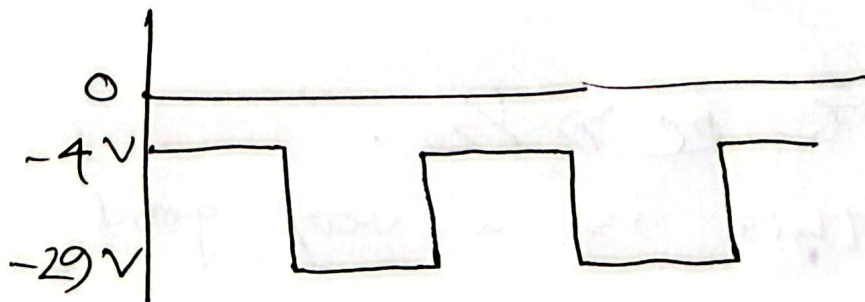
$$\therefore V_i + V_c + V_o = 0$$

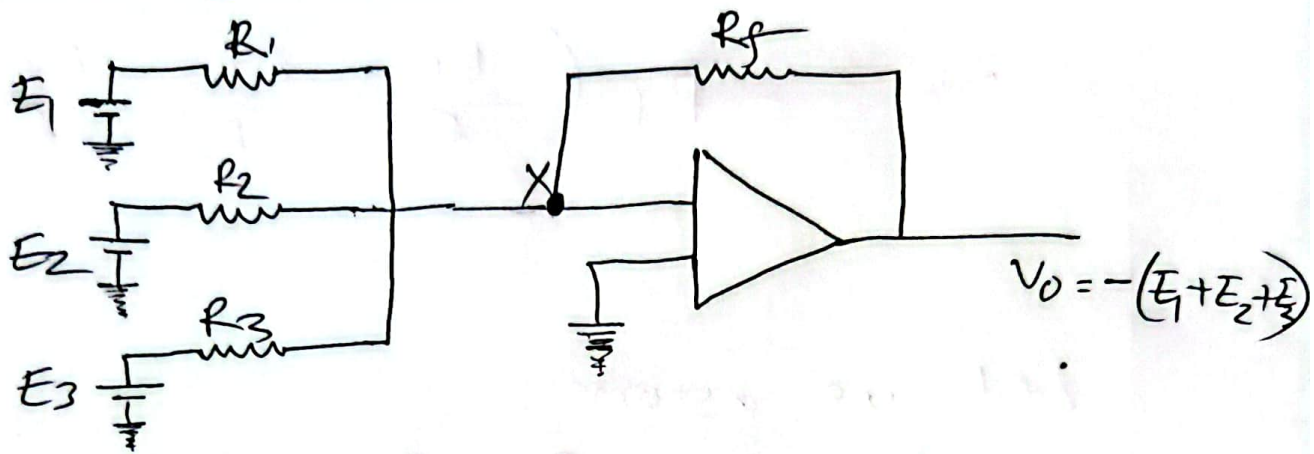
$$\therefore V_o = -V_i - V_c$$

$$= -15 - 14$$

$$= -29V$$

Output waveform:



Inverting Adders CircuitWorking Principle

We know, whenever there is a negative feedback to the Op-Amp then there is a virtual short between inverting and non-inverting terminals. So if one terminal is at ground then another also will be at the ground potential. Hence node X will have 0V.

Now, applying KCL at node X.

$$\begin{aligned}
 I_1 + I_2 + I_3 &= I_f \\
 \Rightarrow \frac{E_1 - 0}{R_1} + \frac{E_2 - 0}{R_2} + \frac{E_3 - 0}{R_3} &= \frac{0 - V_0}{R_f} \\
 \Rightarrow \frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3} &= -\frac{V_0}{R_f} \\
 \Rightarrow V_0 &= -R_f \left(\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3} \right)
 \end{aligned}$$

Output Equation:

$$V_o = -R_f \left(\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3} \right)$$

Let us assume,

$$R_1 = R_2 = R_3 = R_f \quad \text{then}$$

$$V_o = - (E_1 + E_2 + E_3) .$$